Track 1: Genomic Diagnosis of Mosaic Variegated Aneuploidy

Independent candidate-gene interrogation, BUB1B interpretation, and explicit uncertainty framework

Final molecular hypothesis

BUB1B-associated MVA1 is strongly supported, but not molecularly closed: p.Leu737Ter is an established pathogenic loss-of-function allele; p.Asn1002Lys remains a VUS/candidate second allele; phase is unresolved.

VCF records	Candidate-locus records	PASS priority calls	Priority BUB1B calls
5,012,204	1,267	1,260	2

Boundary: research challenge interpretation, not a clinical diagnostic report. Raw genomic data are not reproduced.

Assembly: GRCh38. BUB1B transcript: NM_001211.6.

1. What was actually done

The full local WGS VCF was scanned once and reduced to a pre-specified chromosome-instability/MVA gene space before interpretation. The panel included BUB1B, CEP57, TRIP13, CENATAC, SLF2, SMC5, MAD1L1, BUB1, MAD2L1BP, and CEP192. Technical filtering produced 1,260 PASS variants; presence inside a locus was never treated as evidence of pathogenicity.

WGS (5.01 M records) → 10-gene MVA/mitosis screen → QC (1,260 PASS) → alternative-gene challenge → BUB1B leading signal → phase readiness check → bounded Track 1 conclusion

2. Established evidence vs working hypothesis

Evidence level	Finding	Interpretation
ESTABLISHED	BUB1B chr15:40209701 T>G; c.2210T>G; p.Leu737Ter; 0/1; DP46; GQ99	High-quality nonsense call. ClinVar: Pathogenic for MVA1.
SUPPORTED / UNRESOLVED	BUB1B chr15:40220612 T>G; c.3006T>G; p.Asn1002Lys; 0/1; DP28; GQ99	Technically strong missense call, but remains a VUS/candidate second allele.
UNRESOLVED	Relationship between the two alleles	No BAM/CRAM/FASTQ and no sufficient VCF phase information; cis vs trans cannot be established.
HYPOTHESIS	Biallelic BUB1B dysfunction as the patient's molecular mechanism	Strongly coherent with MVA, but dependent on the unresolved second allele and phase.

ClinVar nuance. p.Leu737Ter is currently classified Pathogenic for MVA1, but the record is based on criteria provided by a single submitter (Labcorp/Invitae), last evaluated October 9, 2024. It is strong external clinical evidence, but not an expert-panel assertion.

3. Why BUB1B remains the leading explanation

The strength is convergence rather than any single score: an established MVA gene, an exact pathogenic loss-of-function allele, a second rare high-quality missense candidate in the same gene, and a phenotype centered on mosaic aneuploidy. The alternative-gene screen was used as a falsification step rather than as a list of genes to force into the diagnosis.

BUB1B alleles (L737* + N1002K?) → BUBR1 reduced/altered → SAC/kinetochore dysfunction → chromosome missegregation → chromosomal instability → mosaic aneuploidy

The question mark is intentional: the mechanistic chain is compelling only if the second allele contributes to BUB1B dysfunction and the alleles form a disease-relevant configuration.

4. Alternative-gene challenge

The screen included established and emerging chromosome-instability genes rather than restricting the search to BUB1B. No alternative configuration identified in the current targeted analysis has comparable combined evidence. This is narrower than claiming that an exhaustive clinical genome analysis excluded every other diagnosis.

5. Phase analysis

A dedicated readiness check found VCF data but no BAM, CRAM, alignment indexes, FASTQ, or local reference FASTA. Read-backed phasing therefore cannot be performed from the available dataset. Two heterozygous calls in one gene are not sufficient to infer trans configuration. Parental segregation or validated long-range/read-level data would be required.

6. Final interpretation

Leading molecular explanation: BUB1B-associated MVA1.

The patient carries a high-quality heterozygous pathogenic BUB1B loss-of-function variant, p.Leu737Ter, together with a second high-quality heterozygous missense variant, p.Asn1002Lys, that remains a VUS/candidate second allele. Phase is unresolved. The appropriate Track 1 conclusion is therefore candidate compound heterozygosity, not confirmed biallelic pathogenic BUB1B disease.

Highest-value evidence still missing

Next evidence	Why it matters
Parental segregation	Establish cis/trans configuration.
Functional N1002K assay	Test BUBR1 abundance, turnover/localization, and checkpoint/kinetochore function.
Cellular phenotype	Measure chromosome alignment, lagging chromosomes, micronuclei, and missegregation.
WT-BUB1B rescue	Provide a causal rescue benchmark if patient-derived cells are available.

7. Independent-analysis safeguards

The interpretation was built from the local case VCF, a pre-specified chromosome-instability/MVA gene panel, technical QC, public clinical evidence, and primary literature. The workflow deliberately separates observed genotype from external evidence and from mechanistic inference. No external participant ranking or submission was used to select, score, or prioritize the reported BUB1B pair.

The distinctive analytical safeguards are: a 10-gene alternative-explanation challenge; explicit separation of established evidence from hypothesis; a documented phase-readiness check; and a genotype-to-mechanism handoff that preserves uncertainty rather than treating the case as molecularly closed.

8. Track 1 → Track 2 handoff

Track 2 should not assume that the genotype has been completely solved. It should ask whether residual BUBR1 function exists and whether a pharmacologic intervention can preserve or compensate for that residual function without worsening chromosome segregation.

Established	Unknown	Track 2 consequence
L737* pathogenic LoF	N1002K functional effect	Do not assume complete BUBR1 absence.
BUB1B-MVA mechanism is biologically coherent	cis vs trans	Drug hypothesis must remain valid under genotype uncertainty.
Missegregation is mechanistically relevant	Residual BUBR1 abundance/localization	Prioritize measurable cellular rescue endpoints, not symptom-only rationale.

9. References and provenance

NCBI ClinVar. NM_001211.6(BUB1B):c.2210T>G (p.Leu737Ter), RCV000641226.9. Pathogenic; criteria provided, single submitter; last evaluated Oct 9, 2024.
Hanks S et al. Constitutional aneuploidy and cancer predisposition caused by biallelic mutations in BUB1B. Nature Genetics. 2004. PMID: 15475955.
Limitations. No parental data; no BAM/CRAM/FASTQ for read-backed phase; no functional assay of N1002K; targeted candidate-gene analysis is not equivalent to exhaustive clinical genome interpretation. Findings are research hypotheses and not for patient care.
AI/data-handling disclosure. OpenAI ChatGPT supported public-literature discovery, synthesis, drafting, methodology/code planning, and targeted review of reduced/candidate-level VCF-derived data. Raw BAM/CRAM and the full VCF are not reproduced in this report. Restricted genomic data should not be redistributed publicly.
AI assistance: OpenAI ChatGPT Plus was used collaboratively for literature synthesis, code development, methodological review, and scientific writing; the account-level "Improve the model for everyone" setting was subsequently disabled.